

6. MASS AND BALANCE

6.1 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Engine Air Cleaner System	D133-2760	27.0	2500	67500
Engine Air Cleaner System	105-85316	22.6	2900	65540
– Installation Provisions	105-85336	9.3	3200	29760
– Particle Separator	105-85321	13.3	2700	35910
Engine Air Cleaner System	105-85318	35.1	2700	94770
– Installation Provisions	105-85334	21.8	2700	58860
– Particle Separator	105-85321	13.3	2700	35910
Engine Air Cleaner System	105-85320	24.0	2700	64800
– Particle Separator Installation	105-85315	18.0	2700	48600
– Enlarged Oil Cooler	105-85312	6.0	2700	16200
Engine Air Cleaner System with Bleed Air Valve	105-84961	23.3	2900	67570
– Installation Provisions		11.3	3200	36160
– Particle Separator		12.0	2700	32400
Engine Air Cleaner System with Bleed Air Valve	105S-85313	(23.3)	(2900)	(67570)
– Installation Provisions	105S-85311	(11.3)	(3200)	(36160)
– Particle Separator	105-85321	(12.0)	(2700)	(32400)
Engine Inlet Deflector Shield	105-62515	0.6	3000	1800
NOTE Above values in parentheses apply to Model S Variants.				

7. SYSTEM DESCRIPTION

The engine air cleaner system is designed to prevent fouling and erosion of rotating components by separating sand and dust particles from the engine intake air. Particle separation is accomplished as the contaminated outside air passes through two panels having several swirl tubes. As the air enters each swirl tube the heavier dirt particles (and water) are thrown outward for collecting while the remaining clean air, located at the eye of the swirl tubes, proceeds towards the engine air intakes. The collected dirt particles are continuously ejected overboard through a bleed air powered scavenge system.

EFFECTIVITY Air Cleaner System D133-2760

The system comprises a square-shaped roof-mounted particle separator, bleed air lines, a bypass control lever with cable assembly, a modified oil cooler air intake and a seal assembly to cover openings in the cowl around the main rotor mast and control rods.

The particle separator houses the swirl tube panels and is fitted with two bypass doors on the forward face. A mechanical control lever, mounted to the RH side of the overhead